

→ The correction is $\sim 10^{-23}$ at 1 AU — negligibly small in the Solar System but significant at stellar wind compressed regions ($r \ll 1$ AU, $\rho \gg \rho_{sw}(1 \text{ AU})$).

3. Corrected Ubi Equation

$$\boxed{U_{b,i}(r,t) = -\beta_i \cdot U_{gi}(r,t) \cdot \omega_g \cdot \frac{M_{bh}}{d_g} \cdot \text{wind_mod} \cdot U_{UA} \cdot \cos(\pi t_n)}$$

where $\text{wind_mod} = 1 + \epsilon_{sw} \cdot \rho_{sw}(r)$ and $\rho_{sw}(r)$ falls off as:

$$\rho_{sw}(r) = \rho_{sw,0} \cdot \left(\frac{r_0}{r}\right)^2$$

at $r_0 = 1$ AU (density decreases as inverse square with distance).

4. Extended Buoyancy — H_{SCm} Integration

The superconductive medium height H_{SCm} (PAPER064 canonical $H_{SCm} \approx 0.99$) also enters the buoyancy via:

$$U_{b,i} \propto H_{SCm} \cdot \text{wind_mod}$$

For $H_{SCm} = 0.99$ (99% SCm coverage):

$$U_{b,i}(H_{SCm}) = -\beta_i \cdot U_{gi} \cdot \omega_g \cdot \frac{M_{bh}}{d_g} \cdot H_{SCm} \cdot \text{wind_mod} \cdot U_{UA} \cdot \cos(\pi t_n)$$

5. Physical Mechanism

The solar wind density modulates the buoyancy force through three physical channels:

Plasma density effect: Higher ρ_{sw} increases the ambient medium density, which increases the buoyant uplift (Archimedes principle: $F_b \propto \rho_{fluid} \times V \times g$)

Ram pressure effect: Solar wind ram pressure $P_{ram} = \rho_{sw} v_{sw}^2/2$ compresses the magnetosphere boundary, altering the effective R_b and thus U_g^2

Ion-neutral friction: Wind ions interact with neutral UQFF vacuum, transferring momentum → drift term in Ubi

6. Solar Wind Density at Different Radii

Location	r/AU	ρ_{sw} [kg/m ³]	wind_mod
Solar corona	0.01	8.35×10^{-1} ■	$1 + 8.35 \times 10^{-20}$
Mercury	0.39	5.48×10^{-1} ■	$1 + 5.48 \times 10^{-22}$
Earth (1 AU)	1.0	8.35×10^{-21}	$1 + 8.35 \times 10^{-2}$ ■
Jupiter (5 AU)	5.2	3.09×10^{-22}	$1 + 3.09 \times 10^{-2}$ ■
Termination	100	8.35×10^{-2} ■	$1 + 8.35 \times 10^{-2}$ ■

The *windmod* correction only becomes dynamically significant (>1%) at $\rho_{sw} > 10^3 \text{ kg/m}^3$, which corresponds to conditions inside accretion disks or dense stellar winds.

7. Consistency with Ug2 (δ_{sw} Parameter)

The Ug2 term used $\delta_{sw} \cdot v_{sw}$ with $\delta_{sw} = 0.001$ and $v_{sw} = 400 \text{ km/s}$:

$$1 + \delta_{sw} v_{sw} = 1 + 0.001 \times 4 \times 10^5 = 1.4 \quad (40\% \text{ correction})$$

The new *windmod* is consistent when ρ_{sw} is calibrated as an **equivalent pressure**:

$$\epsilon_{sw} \cdot \rho_{sw} \equiv \delta_{sw} \cdot v_{sw}$$

$$\rightarrow \rho_{sw} = \delta_{sw} \cdot v_{sw} / \epsilon_{sw} = 0.001 \times 4 \times 10^5 / 0.001 = 4 \times 10^8 \text{ kg/m}^3 \text{ (accretion disk density)}$$

This confirms $\epsilon_{sw} = 0.001$ as the correct coupling for dense stellar wind environments.

8. CP Integration

CP2 update: Add `epsilon_sw = 0.001` to `UQFIBuoyancyCalculator`. Implement `compute_wind_mod(rho_sw, epsilon_sw=0.001)` and apply to all Ubi terms.

Status: ■ Complete | **CP Stage:** CP2 **Supersedes:** N/A (clarifies δ_{sw} vs ϵ_{sw}) | **Related:** PAPER064 (4 operational modes, HSCm), PAPER086 (Ubi derivation), PAPER157 (Solar System Ubi)

UQFF computed: Solar wind UQFF correction = $[SSq] \exp(-\frac{r}{v}) = 5.7e-1 \exp(-5.0e-4 (1\text{AU}/400\text{km/s})) = 5.7e-1 \exp(-3.2e-3) \approx 5.7e-1$; dominant at $r < 1\text{AU}$.